

SDNet: Sandwich Decoder Network for Waterbody Segmentation in Remote Sensing Imagery

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Abstract—Waterbody extraction is essential for monitoring surface changes and supporting disaster response. However, differences in morphology, dimensions, and spectral reflectance make it problematic to segregate waterbodies accurately in remote sensing (RS) photographs. Although self-attention (SA) and convolutional neural networks have demonstrated strong modeling capabilities on RS image recognition, their efficiency and expressiveness still deserve further optimization. In this article, we propose SDNet, an innovative deep learning structure designed to improve segmentation accuracy and scalability in waterbody extraction while minimizing computational costs. SDNet introduces a two-step SA strategy to selectively aggregate spatial features and a sparse-tight pattern fusion module to capture context across various manipulation spaces, enhancing axial and space information fusion. The network's sandwich-shaped decoder structure allows for easy integration with mainstream encoders. Compared to previous influential methods, SDNet exhibits superior sensitivity in detecting small waterbodies and delivers better overall performance. Experiments performed on public benchmark datasets, such as MSRWD, UAVid, and LoveDA, indicate that SDNet exceeds prior state-of-the-art techniques, with a 2–10% increase in mean intersection over union (IoU) and a 3–9% improvement in F1-score. On the MSRWD, SDNet attained an IoU of 94.21% for the water class, with an F1-score of 97.03% and an overall accuracy of 98.87%.

Index Terms—Fusion perception, sandwich decoder, selective attention, semantic segmentation, waterbody extraction.

I. INTRODUCTION

THE rapid advancement of sensor technology and observation platforms has facilitated the acquisition of extensive remote sensing (RS) pictures with diverse temporal and geographical resolutions over the past few years. These data have been extensively used in many downstream tasks, including object recognition [1], change detection [2], land cover mapping [3], flood disaster monitoring [4], and so on. Among these tasks, waterbody segmentation (WS) has consistently garnered attention due to its critical role in vegetation growth analysis, disaster prediction, environmental protection, urban planning, etc. [5], [6], [7]. Therefore, we have chosen it as the focus of our research.

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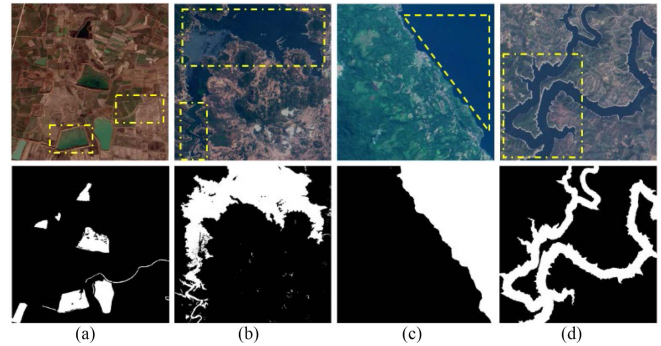


Fig. 1. (a)–(d) Illustration of the challenges encountered in WS. The targets vary significantly in size, reflectance, and shape.

However, accurately and efficiently extracting waterbodies presents various significant challenges. We have analyzed and summarized these difficulties, using specific image examples for illustration, as shown in Fig. 1.

- 1) *Diversity in shape and size*: Waterbodies vary greatly in shape and size, ranging from winding rivers to vast lakes. Their boundaries can be highly irregular and complex.
- 2) *Variations in color and texture*: Waterbodies exhibit significant differences in color and texture due to varying lighting conditions, sediment content, surrounding vegetation, and obstructions.
- 3) *Resolution discrepancies*: Although high-resolution satellite imagery may depict fine-grained spatial details, differentiating waterbodies from other terrestrial features poses significant challenges. The delineation of waterbodies in the photographs may be ambiguous, resulting in water masks that are often shaped by the observer's subjective interpretation.

Early waterbody extraction commonly relied on threshold-based methods, such as single-band thresholding and spectral water indices. The former leverages the low reflectance characteristics of water in the near-infrared (NIR) band to determine a segmentation threshold for distinguishing water from other land cover types. While simple to implement, this method often yields coarse results and is highly susceptible to atmospheric interference, such as clouds and haze. The latter category—spectral water indices—makes use of information from multiple spectral bands and achieves improved segmentation performance at a relatively low cost, leading to widespread adoption in commercial applications. Representative examples include the Normalized Difference Water Index (NDWI) [8] and the Automated Water

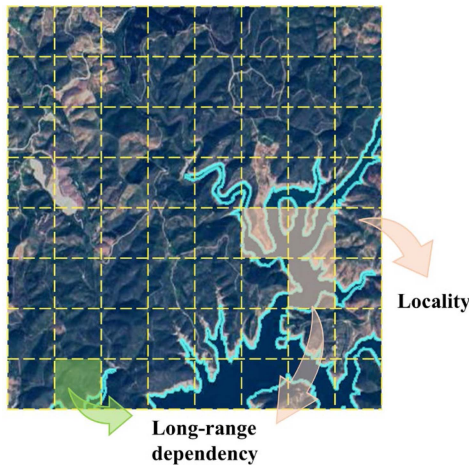


Fig. 2. Two key elements of accurate semantic segmentation, locality and long-range dependencies. Convolutions excel at capturing locality, while SA is proficient at modeling long-range dependencies.

Extraction Index [9]. Although numerous water indices have been proposed, they generally struggle in complex scenarios involving shadows or heterogeneous landscapes. Moreover, most of these methods still require manual threshold adjustment, which becomes impractical for large-scale images and extensive datasets.

In contrast, emerging artificial intelligence (AI)-based image processing methods offer new approaches for tackling the aforementioned issues [10], [11].

Convolutional neural networks (CNNs) are a potent nonlinear representation model for image processing, built on the inductive biases of locality and translational invariance. Convolutional operators capture features by aggregating regional data and stacking layers [12], effectively overcoming the limitations of manually designed feature engineering. Fully convolutional networks (FCNs) have shown strong performance in object extraction tasks [13], [14]. Subsequent several studies have attempted to address the challenges of fixed kernel size and limited receptive field (RF) by introducing strategies such as dilated convolutions [15], [16], depthwise convolutions [17], and deformable convolutions [18], which have also achieved performance improvements in tasks like land cover mapping, change detection, and so on. Despite these advantages, convolutional operators struggle to effectively capture long-range dependencies. Distant pixels cannot communicate directly, thus necessitating many stacked layers to achieve remote information transmission.

In contrast, self-attention (SA) weighted sum excels in identifying global semantic relationships [19]. Vision Transformers (ViTs) [20] have demonstrated superior modeling capabilities across various visual tasks, due to their ability to capture both local patterns and long-range dependencies, as shown in Fig. 2.

In RS image segmentation tasks, we note that several recent methods (i.e., UNetFormer [21] and BuildFormer [22]) combine CNNs and ViTs to leverage their respective strengths, achieving

strong performance in RS image segmentation. However, these methods primarily adjust model capacity by scaling the encoder backbone (e.g., replacing ResNet-18 with ResNet-50 or ResNet-101), which limits flexibility and fine-grained scalability. Motivated by this, we propose introducing adjustable parameters at the decoder side to control model complexity more directly, enabling scalable and flexible deployment. Our design is also modular, making it easy to integrate with other components.

Focusing on the topic of waterbody extraction, we highlight that prior deep neural network studies have effectively employed encoder–decoder architectures to learn high-level semantic features and achieve pixel-level upsampling precision [5], [23], [24], [25]. Among works emphasizing decoder design, Hertel et al. [26] introduced a probability layer at the end of each decoder block to model weight distributions using normal distributions and estimate posterior parameters via Bayes' theorem. This approach offers theoretically grounded prediction intervals and high architectural flexibility. However, their experiments show limited improvements over baselines in complex scenes, particularly under class imbalance conditions. Zhong et al. [27] proposed a multilevel transformer module to unify multiscale features through arithmetic averaging and Hadamard product. While this fusion strategy proved effective, it incurred frequent shape adjustments, leading to information loss and high computational overhead. Dai et al. [28] designed a dual-branch multiscale aggregation unit and a multiscale fusion unit, which extracted hierarchical features via convolutions with varying sampling rates and fused high-level semantics with low-level visual cues. Despite promising results, their approach lacked an effective tradeoff between segmentation performance and model complexity, limiting its deployment on resource-constrained hardware.

Most of the aforementioned methods struggle to strike an effective tradeoff between water segmentation accuracy and computational cost. While some models achieve improved segmentation performance, they often incur significantly higher computational overhead. Conversely, when these models are optimized for lightweight design, their segmentation performance tends to degrade, resulting in reduced accuracy metrics. Achieving a balance between accuracy and efficiency remains a promising and important research direction.

Some recent efforts have attempted to address this issue, for example, by replacing monolithic convolutions with layerwise or dilated convolutions, e.g., MSLANet [28], or by substituting global attention with window-based attention mechanisms, e.g., CSWin [29].

Building upon these ideas, our work further aims to improve model flexibility and scalability through both theoretical analysis and empirical validation. In this article, we propose a two-step self-attention (TSSA) method and a sparse-tight pattern fusion (STPF) module. The TSSA method first removes redundant calculations in SA through a pooling-based selection strategy, followed by key–value pair matching and elementwise weighted summation. The STPF module uses both large sparse and small tight kernels to capture features at different spatial scales, followed by channel partitioning and recombination for

axial information fusion, yielding strong performance in segmentation tasks.

This work's primary contributions can be encapsulated as follows.

- 1) SDNet is designed for WS, with each submodule of the decoder structured in a "sandwich" fashion. The lower layer (TSSA) method improves the transformer's computation process and has been proven effective both theoretically and experimentally.
- 2) The upper layer (STPF) module integrates both spatial and channel dimension information by employing strategies such as large and small kernel convolutions and partitioning-recombination techniques. The dual-branch interactive structure increases the model's expressive power.
- 3) SDNet exhibits enhanced mean intersection over union (mIoU) while decreasing floating-point operations (FLOPs) on our proprietary RS dataset. Experimental findings on the MSRWD, UAVid, and LoveDA datasets demonstrate that SDNet surpasses current state-of-the-art (SOTA) techniques.

The rest of this article is organized as follows. Section II examines pertinent literature on SA optimization, multikernel multibranch convolution, and semantic segmentation of waterbody. Section III presents a detailed and comprehensive elucidation of SDNet. Section IV delineates ablation investigations and comparison experiments on benchmark datasets to substantiate the efficacy of SDNet. Finally, Section V concludes this article and outlines future directions for improvement.

II. RELATED WORKS

A. SA-Based RS Image Segmentation

Among various deep learning algorithms, Transformer [19] and SA excel in long-sequence modeling, with ViT [20] being the pioneering effort in computer vision, yielding strong results across various tasks. Subsequently, numerous studies have been proposed to make it more efficient. Swin Transformer [30] introduced sliding window attention for localized feature extraction and fusion. Pyramid ViT [31] reduced sequence length progressively with network depth. SegFormer [32] enhanced computational efficiency by replacing SA with multilayer perceptrons (MLPs) in the decoder. In addition, methods like CSWin Transformer [29] and CrossFormer [33] have adjusted key-value pair matching to better balance performance and computational cost.

However, transformers are often burdened by high parameter counts, computational demands, and weakened local association modeling. Building on these insights, we advocate for window-based feature extraction combined with reduced redundant computation to achieve a balanced performance.

B. CNN-Based RS Image Segmentation

Convolution [13] is effective at capturing local features, largely determined by the size and shape of the convolutional

kernels. To aggregate multilevel RF information from a single feature map, some works employed parallel multibranch structures with various block [34] or strip convolutions [35], [36], or varied dilation rates [37], [38]. In addition, encoder-decoder networks, i.e., Unet [39], U2Net [40], efficiently acquire multiscale attributes using upsampling and downsampling processes. Building on this, Wnet [2] proposed Double-Unet, with two upsampling paths for superpixel generation and change-pixel discrimination. HRNet [41] enhanced cross-layer interactions through dense feature fusion. Following the introduction of ViT, ConvNeXt [42] adopted convolution operators to mimic Swin Transformer [30] and demonstrated that large kernels could outperform shifted window attention, while SegNeXt [36] achieved similar gains with larger kernel strip convolutions.

Despite these achievements, optimizing CNN-based methods remains a worthwhile direction of research. For instance, introducing a combination of layerwise sparsity and compact awareness can effectively capture spatial features under each pattern. Moreover, feature dimension splitting and reorganization may contribute to enhancing information fusion and segmentation performance.

C. Waterbody Segmentation

The FCN [43] pioneered deep learning for semantic segmentation, followed by numerous CNN-based methods for RS tasks [13], [14], [18], [44]. The limitations of convolutions led to increased interest in SA mechanisms, with CNN-ViT hybrids showing advantages in pixel-level image segmentation [21], [22], [23], [45]. Early WS approaches introduced cascade CNNs [38] with upsampling pyramids to reduce information loss and conditional random fields for computational efficiency. deep stacked ensemble model (DSEM) [46] used parallel CNN models to reduce variance, while SG-WaterNet [47] applied graph neural networks to compress superpixel data. Dai et al. [28] propose parallel multipath dilated convolutions to capture spatial information. NT-Net [27] introduces a multilevel ViT to fuse and express convolutional pooling outputs at different resolutions. Waterformer [23] leverages a dual-stream CNN and a cross-layer attention mechanism for feature modeling, effectively capturing multilevel waterbody features. In addition, it incorporates a subpixel upsampling module to generate high-resolution image representations. Wei et al. [48] proposed MLFMNet, a CNN-based decoder designed for aerial RS image segmentation. Integrated with an encoder, it forms a multilevel feature mining network framework. The decoder incorporates the learnable fusion module (LFM) and feature reconstruction module (FRM) modules to effectively fuse and reconstruct information across different feature levels. Nonetheless, a slight limitation of this work is that the improvement in segmentation performance comes at the cost of significantly increased computational overhead. Yang et al. [49] proposed WatNet, which, similar to our work, focuses on the subdomain of waterbody extraction and emphasizes improvements on the decoder side. The decoder enhances segmentation accuracy in complex environments through convolution-based global multiattention

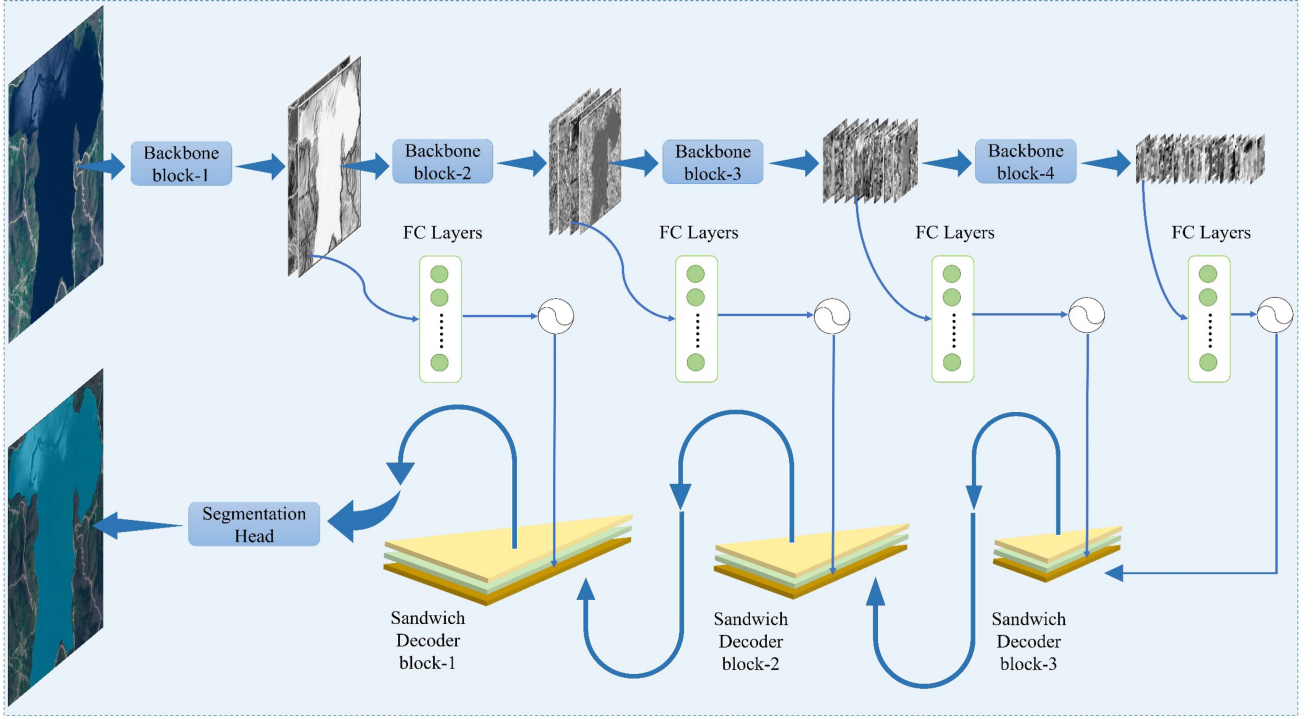


Fig. 3. Overview of the SDNet, consisting of a feature aggregation branch and a semantic comprehension branch.

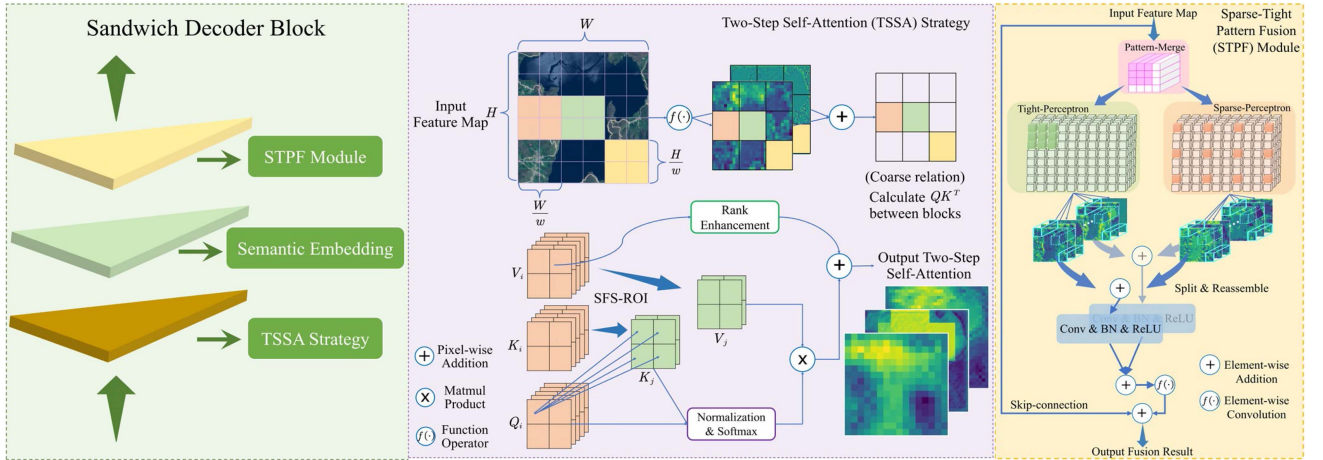


Fig. 4. Details about the sandwich decoder, with the lower layer TSSA strategy, the middle layer original feature semantic embedding, and the upper layer STPF module.

fusion, a waterbody forward network, and edge focus attention modules.

Most mentioned methods follow a strategy of optimizing the entire network, improving both the encoding and decoding processes. Some methods [26] focus solely on decoding, yet show constrained effectiveness. Methods [23], [27] leveraging SA mechanisms are often restricted to providing input via compressed concatenations at different levels. This work theoretically addresses the redundancy in SA computations. While CNN-based approaches are well established, direct feature concatenation risks diluting the learning focus. We propose a hybrid sequential–parallel strategy with grouped information

summation to mitigate this issue. To the best of our knowledge, no prior work has explored eliminating SA computational redundancy with theoretical analysis in the topic of WS. Furthermore, we propose a novel decoder-only design, enabling seamless integration with evolving mainstream encoder backbones as a lightweight module for large-scale RS models.

III. METHODOLOGY

As depicted in Figs. 3 and 4, the proposed SDNet consists of a downsampling feature aggregation branch and an upsampling semantic comprehension branch. SDNet derives its name from

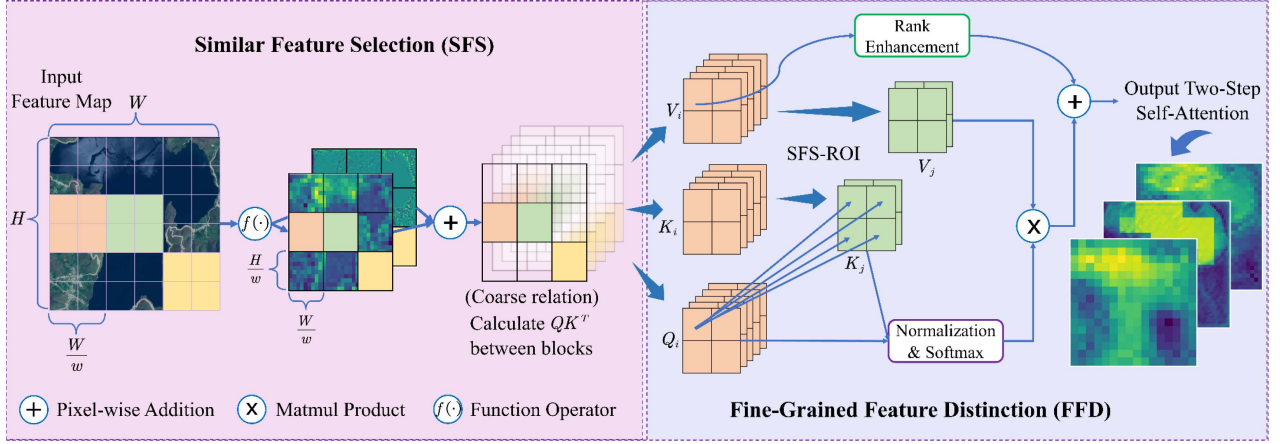


Fig. 5. Illustration of the TSSA strategy.

the layered architecture employed in its decoder design. The subsequent sections include a comprehensive description of each component.

A. Two-Step Self-Attention

The lower level feature maps, after several aggregation steps, contain rich semantic information. To reduce redundant computations, we propose a TSSA strategy that first selects similar features and then refines them.

1) *Similar Feature Selection (SFS)*: Directly computing the attention between each element on a feature map results in complexity of $O(n^2)$, which is computationally expensive. To balance accuracy and efficiency in WS, we adopt a region-of-interest (ROI) strategy. First, 1×1 convolutions and splitting operations are applied to the original input feature map, generating query (Q), key (K), and value (V)

$$\begin{aligned} Q &= W^q M_{\text{in}} \\ K &= W^k M_{\text{in}} \\ V &= W^v M_{\text{in}}. \end{aligned} \quad (1)$$

Then, we set nonoverlapping windows of size $w \times w$ to aggregate regional information across the entire feature map, using average and max pooling. Moreover, the coefficients of their linear combinations are trainable. In Fig. 5, we use the function operator to represent; here, only pooling is considered to reduce the amount of calculation and parameter. To achieve better accuracy, a more comprehensive feature compression module could be employed or, alternatively, made learnable.

The pooled query, key $\in \mathbb{R}^{C \times H/w \times W/w}$ are used to compute semantic similarity among windows

$$\begin{aligned} Q^c &= \lambda_{Q1} P_{\text{avg}}(Q) + \lambda_{Q2} P_{\text{max}}(Q) \\ K^c &= \lambda_{K1} P_{\text{avg}}(K) + \lambda_{K2} P_{\text{max}}(K). \end{aligned} \quad (2)$$

From the similarity matrix, we select the top- k values as ROI. A larger value of k indicates a more significant role in the subsequent weighted summation calculations. For each nonoverlapping window w_i , we focus only on the k most relevant

ones $w_1, w_2, \dots, w_k$ for the subsequent fine-grained feature distinction

$$\text{ROI}(M_{\text{in}}) = \text{Top}k \left(Q^c (K^c)^T \right). \quad (3)$$

SFS has already taken spatial feature correlations into account through the function operator and cross-channel feature similarity through the QK^T matrix product. Consequently, the selected k candidate regions are most representative. Regions with low ROI values contribute minimally to the weighted sum in SA computations.

2) *Fine-Grained Feature Distinction (FFD)*: After selecting the k candidate ROIs for all regions, the original feature tensor is mapped to all query, while the candidate regions are mapped to key-value pairs. Here, we save a lot of computational redundancy.

The original map retains all information, while the ROI map highlights the most critical information. We then perform multihead attention to distinguish finer differences

$$\text{MSA}(Q_i, K_j, V_j) = \text{softmax} \left(\frac{Q_i K_j^T}{\sqrt{d_k}} \right) V_j. \quad (4)$$

Considering the window size $w \times w$, the comparison of global multihead self-attention (MSA) and ROI-MSA in terms of computational complexity for a feature tensor of size $H \times W \times C$ is as follows:

$$O(\text{G-MSA}) = 4HWC^2 + 2(HW)^2C \quad (5)$$

$$\begin{aligned} O(\text{ROI-MSA}) &= 4HWC^2 + \left(\frac{HW}{w^2} \right)^2 C + 2kw^2HWC \\ &\geq 4HWC^2 + 3k^{\frac{2}{3}}(HW)^{\frac{4}{3}}C. \end{aligned} \quad (6)$$

Equality holds when $HW = kw^6$, minimizing computational cost. By setting appropriate parameters, this approach removes a significant amount of redundant computation, balancing segmentation accuracy and efficiency, and reducing the original quadratic complexity from $O((HW)^2)$ to $O((HW)^{4/3})$ in the best case.

After computing the fine-grained attention, we apply pattern-wise convolutions to enhance the rank of the feature tensor [17], improving the network's ability to represent and differentiate patterns

$$M_{\text{out}} = \text{ROI} - \text{MSA}(M_{\text{in}}) + \text{PWC}(V). \quad (7)$$

Fig. 5 illustrates the whole execution process of the TSSA strategy, which first performs SFS, followed by FFD.

B. Sparse-Tight Pattern Fusion

Shallow feature maps retain detailed information, while deeper maps capture higher level semantic abstractions. The challenge in pixel-level segmentation is describing features within arbitrary irregular regions. Our approach combines local perception with surrounding context.

We design a serial-parallel pattern fusion neural network, incorporating prior knowledge from digital image processing, such as locality and RFs. Locality limits convolutional perception to the kernel size, while RFs account for distant pixel relationships that are crucial for accurate segmentation.

Our proposed network employs a bottleneck structure for feature compression and reuse. Given a 2-D input feature map $M_{\text{in}} \in \mathbb{R}^{C \times H \times W}$, a convolution-normalization-pooling cuboid is first applied to merge cross-pattern information. Next, we integrate a parallel convolution module, where the upper branch employs 3×3 local perceptron to capture compact features, while the lower branch uses 7×7 sparse perceptron to extract large-scale environmental semantics. The initial pattern fusion block applies a single operation across all channels of the feature tensor, fully integrating multimodal information. In contrast, the subsequent parallel perception blocks operate layer by layer, aiming to preserve modality diversity as much as possible. Finally, 1×1 convolutions are employed to compress the channels for better feature reuse

$$M_{\text{pre}} = \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(M_{\text{in}}))) \quad (8)$$

$$M_{\text{loc}} = \text{Conv}_{1 \times 1}(\text{Conv}_{3 \times 3}(M_{\text{pre}})) \quad (9)$$

$$M_{\text{sur}} = \text{Conv}_{1 \times 1}(\text{DilatedConv}_{7 \times 7}(M_{\text{pre}}, r)). \quad (10)$$

The dilation rate r adjusts the RF size, influencing the ability to absorb semantics from the surrounding environment

$$\text{RF} = (k - 1)(r - 1) + k. \quad (11)$$

In (11), k represents the kernel size of 2-D convolution. Dilated convolutions capture distant spatial features, but excessive dilation can lead to overpadding, while small dilation limits long-range dependence modeling. We set $r = 3$ and $k = 7$ for balance. In practical tasks, this is an adjustable parameter that can be modified based on experimental results.

The feature maps are split and reorganized along the channel dimension to facilitate multiple recognition patterns

$$M_{\text{loc}} = M_{\text{loc}}^1 \cup M_{\text{loc}}^2 \quad (12)$$

$$M_{\text{sur}} = M_{\text{sur}}^1 \cup M_{\text{sur}}^2 \quad (13)$$

$$M_{\text{attn}}^1 = M_{\text{loc}}^1 + M_{\text{sur}}^1 \quad (14)$$

$$M_{\text{attn}}^2 = M_{\text{loc}}^2 + M_{\text{sur}}^2. \quad (15)$$

Here, we use summation instead of concatenation, aligning the feature dimensions to avoid information confusion.

A standard 3×3 convolution fuses information across these two branches

$$M_{\text{attn}}^i = \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(M_{\text{attn}}^i))) \quad i = 1, 2. \quad (16)$$

Finally, the resulting dual-branch feature maps, M_{attn} , are merged by elementwise summation and subsequently upsampled to recover the original channel dimensions. An identity mapping from the input is incorporated, yielding the final fused semantic representation

$$M_{\text{out}} = M_{\text{in}} + \text{Conv}_{1 \times 1}(M_{\text{attn}}^1 + M_{\text{attn}}^2). \quad (17)$$

Fig. 6 illustrates the intricate structure and workflow of the STPF module, which begins with pattern merge and sparse-tight perception, followed by modality splitting, reassembly, and refinement through summation.

C. Network Structure and Training Strategy

1) *Feature Aggregation Branch*: For the input RS image, we apply four-stage downsampling to obtain multiscale representations. At each stage, mainstream networks, such as ConvNeXt [42] (CNN-based) or SegFormer [32] (ViT-based), are used for feature extraction backbone. Outputs from each stage are both fed into the next stage and encoded through fully connected layers. These vectors are then reshaped into the middle layers of the sandwich decoder, preserving much of the original image information.

2) *Semantic Comprehension Branch*: Segmentation benefits from multilevel information, which the sandwich decoder excels at capturing. The decoder has three layers: 1) the bottom layer uses a TSSA strategy to capture deep connections and reduce redundant computations; 2) the middle layer retains significant raw information from the feature embedding (FE); and 3) the top layer, the STPF module, integrates features across layers to improve performance.

3) *Loss Calculation Strategy*: Loss function quantifies the disparity between predicted segmentation results and ground truth (GT) labels.

Cross-entropy loss works well in most cases but tends to bias toward the background when foreground pixels are sparse, leading to poor results

$$L_{\text{ce}} = - \sum_{c=1}^M y_c \log(p_c). \quad (18)$$

In (18), M represents the number of classes, y_c designates a one-hot vector, and p_c signifies the probability that a sample is classified as belonging to class- c .

Dice loss prioritizes foreground regions by assigning larger gradients to positive samples

$$L_{\text{dice}} = 1 - \frac{2|X \cap Y|}{|X| + |Y|}. \quad (19)$$

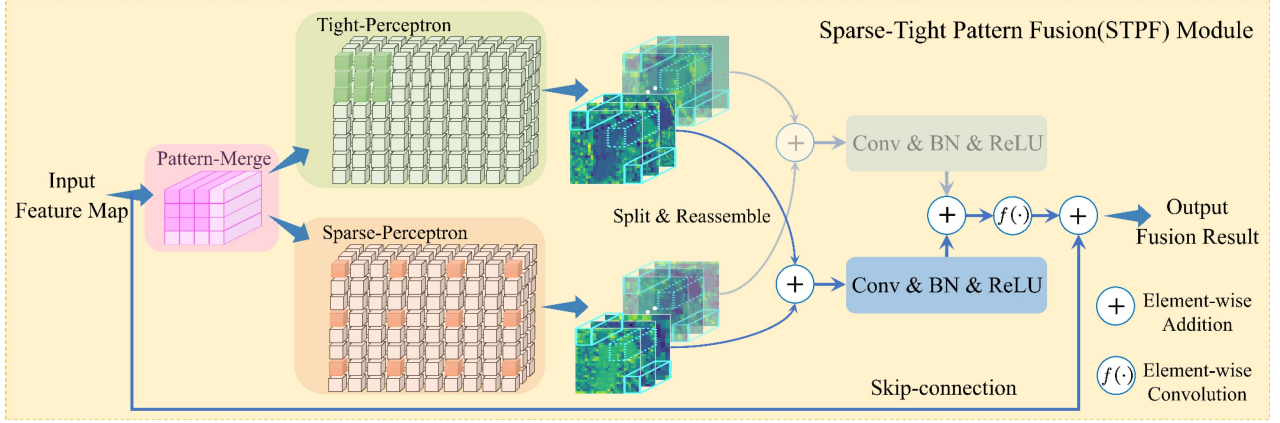


Fig. 6. Illustration of the STPF module.

In (19), X signifies the GT label, whereas Y denotes the predicted result.

In this work, we combine both loss functions with a predefined weighting coefficient k , selecting the optimal setting based on practical performance. The total loss is given by

$$\text{Loss} = L_{ce} + k \times \left(1 - \frac{\text{epoch}_{\text{cur}}}{\text{epoch}_{\text{all}}}\right) \times L_{\text{dice}}. \quad (20)$$

Here, k is a nontrainable parameter set based on practical scenarios. In addition, a loss decay strategy is employed, where $\text{epoch}_{\text{cur}}$ denotes the current iteration number, and $\text{epoch}_{\text{all}}$ specifies the aggregate number of iterations required. This linear decay strategy typically leads to better optimization.

IV. EXPERIMENTS AND RESULTS

In this section, experimental datasets, information regarding the implementation, and assessment metrics are presented for the first time. In the following step, we carry out ablation research on the suggested approach, and then, we proceed to perform comparison tests on various datasets. In the last step, the outcomes and weaknesses of the experiment are examined and described in further depth.

A. Dataset

We trained and evaluated SDNet using the SIWB, MSRWD, UAVid, and LoveDA datasets, which were derived from a variety of sensors, in order to validate its scalability and resilience. The details of these datasets are as follows.

- 1) *SIWB*: This custom RS dataset, captured by the Sentinel-2 satellite, comprises 4083 images with two labels (background—0, waterbody—1). Segmentation challenges stem from large variations in image sizes and severe class imbalance in some instances.
- 2) *MSRWD* [23]: A high-resolution dataset designed for waterbody extraction, MSRWD, contains 660 images with two labels (nonwaterbody—0, waterbody—1). The key difficulty lies in the variation of waterbody sizes, shapes, and reflectance.

- 3) *UAVid* [50]: UAVid is a drone-view segmentation dataset of 300 high-resolution urban street scene images with eight classes (clutter—0, building—1, road—2, tree—3, vegetation—4, moving car—5, static car—6, human—7). Challenges include large spatial dimensions, rich details, and diverse viewpoints caused by temporal variations.
- 4) *LoveDA* [51]: A satellite RS dataset from Google Earth Engine, LoveDA, comprises 5987 images from three cities (Nanjing, Changzhou, and Wuhan), each with a resolution of 1024×1024 pixels at 0.3 m. It includes seven classes (background—0, building—1, road—2, water—3, barren—4, forest—5, agriculture—6). Segmentation challenges arise from the variability across different sampling regions.

B. Experimental Settings

1) *Implementation Details*: All neural networks were implemented utilizing the PyTorch framework and deployed on a single NVIDIA RTX 3080Ti 12-GB GPU.

To fine-tune parameters and accelerate model convergence, we employed the AdamW optimizer along with a cosine annealing learning rate strategy for training. The initial learning rate γ was not fixed but adjusted based on the specific performance of each network. Other hyperparameters were set to their usual default values: $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\varepsilon = 1e - 8$.

For the SIWB dataset, we preprocessed the images by removing those with very low resolution or highly imbalanced positive and negative samples. In addition, we manually corrected certain image masks for better fitting. To further enhance performance, data augmentation techniques were utilized, including random flips, small rotations, and brightness and contrast modifications.

Each image was pasted onto a white background of fixed dimensions (e.g., 512×512 pixels) before segmentation, and the output result was cropped back to the original image area to prevent distortion. Finally, the data were partitioned into two parts according to certain random rule: 80% for training and 20% for validation. The former is employed for

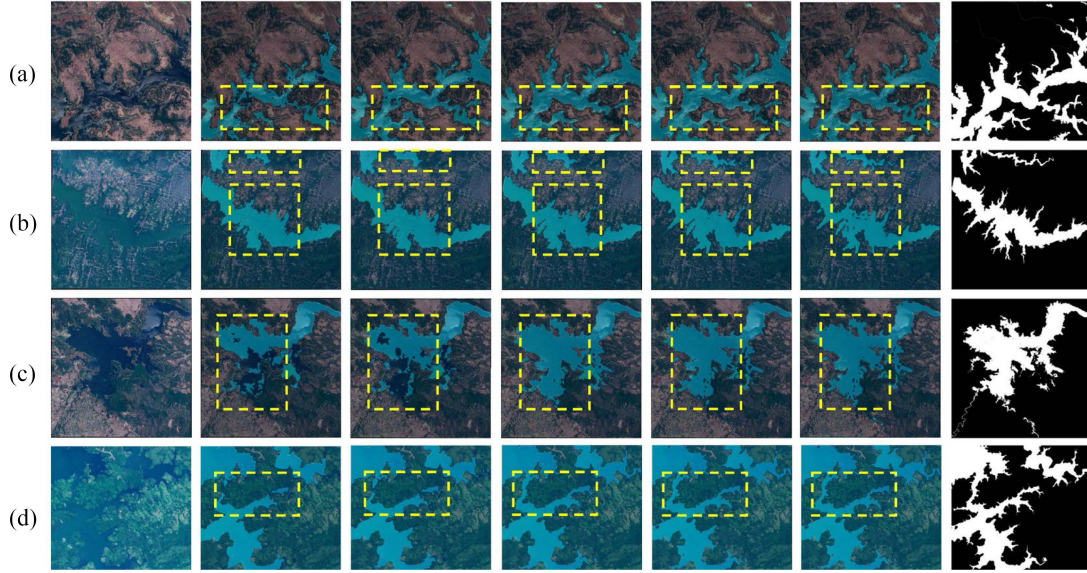


Fig. 7. (a)–(d) Visual analysis of WS results on SIWB. Each row corresponds to an image in the dataset. From left to right, the columns represent the following: Image, SDNet-O, SDNet-T, SDNet-M, SDNet-TM, SDNet-TMF, and label.

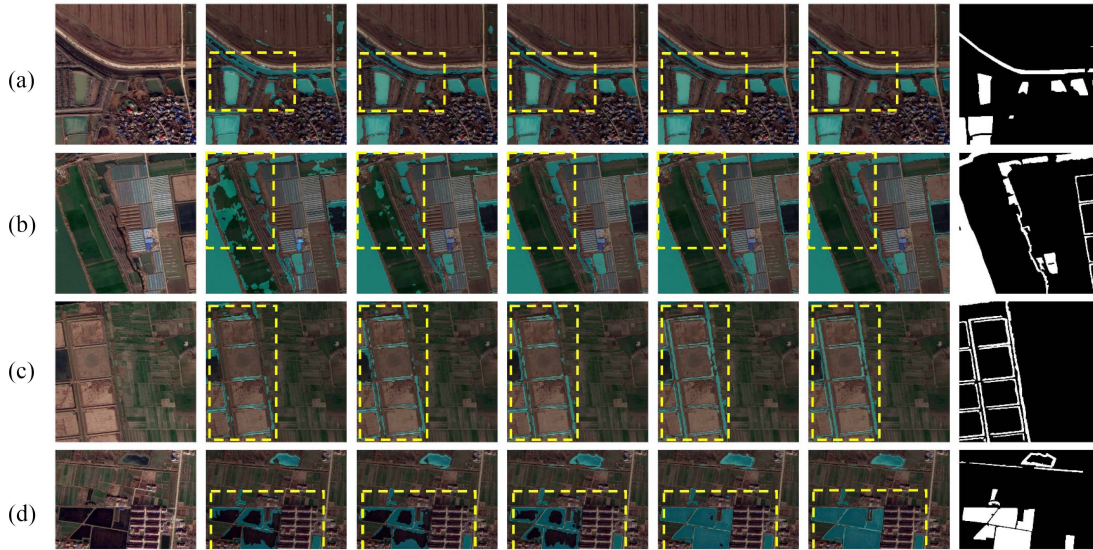


Fig. 8. (a)–(d) Visual analysis of WS results on MSRWD. Each row corresponds to an image in the dataset. From left to right, the columns represent the following: Image, SDNet-O, SDNet-T, SDNet-M, SDNet-TM, SDNet-TMF, and label.

model fitting, while the latter is utilized for assessing model performance.

For the MSRWD, UAVid, and LoveDA datasets, the image-mask pairs were randomly cropped to dimensions of 512×512 pixels. All experiments were executed for 100 epochs to guarantee a fair comparison.

2) *Evaluation Metrics*: This study employs assessment measures that analyze model performance in terms of segmentation accuracy and model representation ability. The former includes mIoU and overall accuracy (OA), as defined in (21) and (22), respectively. Model representation ability is measured by the

number of parameters and FLOPs

$$\text{mIoU} = \frac{1}{n_{\text{cls}}} \sum_i \frac{n_{ii}}{p_i + \sum_j n_{ji} - n_{ii}} \quad (21)$$

$$\text{OA} = \frac{\sum_i n_{ii}}{\sum_i p_i}. \quad (22)$$

Here, n_{ij} denotes pixels of class- i predicted as class- j , p_i signifies the total number of pixels for class- i , and n_{cls} indicates the number of target classes inside the dataset. Since waterbody extraction is the primary focus, we also report the intersection

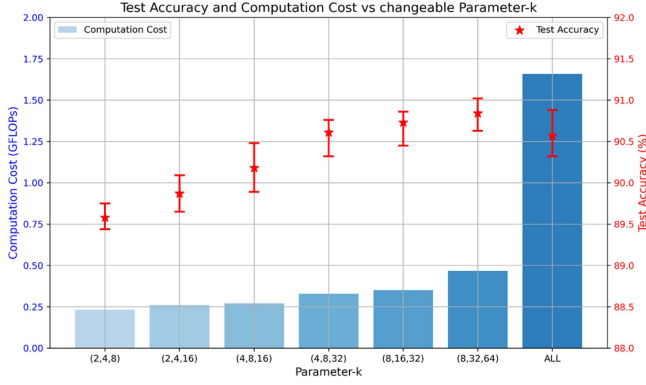


Fig. 9. Comparison experimental results on the waterbody dataset with varying parameter k .

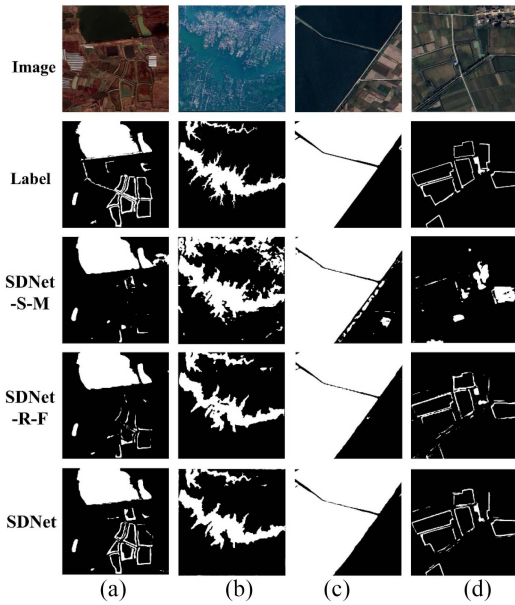


Fig. 10. (a)–(d) Ablation study visualization: Contributions of SFS and FFD to TSSA module.

over union (IoU), precision, recall, and $F1_{\text{score}}$ specifically for the waterbody class, as defined in the following equations:

$$\text{IoU} = \frac{n_{ii}}{p_i + \sum_j n_{ji} - n_{ii}} \quad (23)$$

$$\text{Precision} = \frac{n_{ii}}{n_{ii} + \sum_{j,j \neq i} n_{ji}} \quad (24)$$

$$\text{Recall} = \frac{n_{ii}}{n_{ii} + \sum_{j,j \neq i} n_{ij}} \quad (25)$$

$$F1_{\text{score}} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (26)$$

Assessing accuracy and recall for an individual class yields enhanced understanding of model efficacy. Precision counts the proportion of accurate forecasts among all predicted instances, whereas recall measures the proportion of accurate predictions in

relation to the total number of actual occurrences for that class. The harmonic mean of recall, precision, and $F1_{\text{score}}$ offers a more complete picture of the model.

C. Ablation Study

We undertook many ablation experiments on two waterbody datasets, SIWB and MSRWD, to validate each component in the proposed network. The quantitative results were presented in Tables I and II, and the visualization comparison was shown in Figs. 7 and 8.

We identify the original SegFormer-T encoder, coupled with a stepwise upsampling decoder, as SDNet-O. The TSSA technique, used as the bottom layer of the sandwich decoder block, is designated as SDNet-T, while the STPF module, employed as the top layer of the block, is designated as SDNet-M. The integration of TSSA with STPF for feature augmentation yields SDNet-TM. Ultimately, with the integration of FE, the full model, referred to as SDNet, is denoted SDNet-TMF.

1) *Effect of the TSSA Strategy*: The segmentation results before and after incorporating TSSA are compared in second and third columns of Figs. 7 and 8. The network’s improved capability to capture long-range connections is evident, thanks to the elementwise SA mechanism.

Quantitative results in Tables I and II further demonstrate TSSA’s effectiveness. On the SIWB dataset, $\text{IoU}_{\text{water}}$ increased by 1.87%, $F1_{\text{score}}_{\text{water}}$ by 1.09%, and OA by 0.5%. On the MSRWD dataset, $\text{IoU}_{\text{water}}$ improved by 1.51%, $F1_{\text{score}}_{\text{water}}$ by 0.84%, and OA by 0.28%. These gains were achieved without significant changes in model size or computational cost, due to the efficient TSSA mechanism that reduces redundancy and enhances performance.

We also conducted a series of parameter tuning experiments on the TSSA strategy of the model, and the quantitative results are shown in Fig. 9. The x -axis represents the parameter k values of different layers of the decoder (i.e., the number of regions of interest). The y -axis is divided into two parts: the left-hand side shows the computational cost, corresponding to the bar chart, while the right-hand side displays a scatter plot representing the average accuracy and the upper and lower bounds over ten tests. “ALL” denotes the strategy without the TSSA selection, where the model computes mutual attention for all elements. From the figure, it can be seen that the selection strategy is more effective in establishing associations between RS pixels, with lower computational costs. When larger k values are used, it can even outperform the full association strategy. This is because, by removing redundancy, we achieve an effect similar to random dropout, effectively preventing the model from overfitting.

2) *Effect of the STPF Module*: The segmentation results with and without STPF are compared in second and fourth columns of Figs. 7 and 8. Incorporating STPF markedly improves water extraction accuracy, effectively identifying both large waterbodies and narrow rivers. As illustrated in third and fourth columns, STPF demonstrates superior robustness and accuracy over TSSA, particularly in handling variations in reflectance and resolution.

TABLE I
ABLATION STUDY OF EACH COMPONENT ON SIWB

Method	IoU _{water} (%)	Precision _{water} (%)	Recall _{water} (%)	F1-score _{water} (%)	OA (%)	Params (M)	FLOPs (G)
SDNet-O	84.13	94.26	88.67	91.38	96.02	3.40	0.42
SDNet-T	86.00	95.90	91.16	93.47	96.52	3.70	0.42
SDNet-M	86.85	95.63	90.44	92.96	96.74	6.79	6.44
SDNet-TM	87.52	95.77	91.04	93.35	96.91	7.08	6.45
SDNet-TMF	87.99	95.72	91.65	93.64	97.05	7.28	6.57

TABLE II
ABLATION STUDY OF EACH COMPONENT ON MSRWD

Method	IoU _{water} (%)	Precision _{water} (%)	Recall _{water} (%)	F1-score _{water} (%)	OA (%)	Params (M)	FLOPs (G)
SDNet-O	89.36	95.51	93.28	94.38	97.89	3.40	0.42
SDNet-T	90.87	95.48	94.96	95.22	98.17	3.70	0.42
SDNet-M	91.99	96.87	94.80	95.83	98.43	6.79	6.44
SDNet-TM	92.66	97.06	95.33	96.19	98.56	7.08	6.45
SDNet-TMF	93.04	97.27	95.61	96.43	98.65	7.28	6.57

Tables I and II summarize the quantitative results. Despite an increase in parameters and computational cost due to the dual-branch modeling and reassembly strategy, our approach remains lightweight and efficient compared to SOTA water extraction networks. On the SIWB dataset, IoU_{water} increased by 2.72%, F1-score_{water} by 1.58%, and OA by 0.72%. On the MSRWD dataset, IoU_{water} improved by 2.63%, F1-score_{water} by 1.45%, and OA by 0.54%. The performance gains are attributed to the comprehensive fusion of information from diverse RFs and feature modalities.

3) *Effect of FE*: As the resolution of the feature map diminishes and the number of channels escalates during down-sampling, the direct integration of original picture data into the sandwich decoder results in layer misalignment. To address this, we introduced an MLP for semantic encoding and feature alignment. Although this addition showed minimal visual improvement in water segmentation, quantitative results indicated a notable performance gain. Specifically, on the SIWB dataset, IoU_{water} improved from 87.52% to 87.99%, F1-score_{water} from 93.35% to 93.64%, and OA from 96.91% to 97.05% (shown in Table I). On the MSRWD dataset, IoU_{water} increased from 92.66% to 93.04%, F1-score_{water} from 96.19% to 96.43%, and OA from 98.56% to 98.65% (shown in Table II).

To verify the effectiveness of each component in the TSSA module, we designed additional ablation experiments based on the following logic.

- 1) *Removing SFS*: We replace the SFS module with a mechanism that randomly selects matching key-value pairs, which are then fed into the FFD. We hypothesize that random selection should lead to lower accuracy compared to the targeted filtering of SFS.
- 2) *Removing FFD*: After SFS extracts relevant regions, we feed them into an MLP instead of FFD. Since MLPs

lack the dual spatial and channel-level perception of SA mechanisms, we expect a performance drop.

The quantitative results of these ablation experiments are shown in Tables III and IV. In Tables III and IV, SDNet is the original model, SDNet-R-F is Test 1 (SFS removed), and SDNet-S-M is Test 2 (FFD removed). The ablation results are consistent with our expectations, further supporting the rationality of our network design. We also present several visual comparisons, as shown in Fig. 10.

D. Comparison With Prior SOTA Methods

We compared our proposed technique against several previous SOTA image segmentation networks, including Deeplabv3+ [37], U2Net [40], SwinUnet [52], Crossformer [33], Segformer [32], Unetformer, CSwin [29], ConvNeXt [42], and Seaformer [53], highlighting the effectiveness and efficiency of our proposed method.

Furthermore, although our method is designed for WS, we aim to extend its applicability to more complex instance segmentation tasks. To validate its generalization capability, we conducted experiments not only on two water segmentation datasets, SIWB and MSRWD, but also on two multiclass segmentation benchmarks, LoveDA and UAVid. The quantitative comparison experimental results are recorded in Tables V, VI, IX, and X, while qualitative comparisons are presented in Figs. 11, 14, and 15.

1) *Comparative Experiments on Waterbody Datasets (SIWB and MSRWD)*: We assessed the resilience and generalization capabilities of SDNet by training advanced segmentation networks on the SIWB and MSRWD waterbody datasets, yielding detailed quantitative comparisons. As shown in Tables V and VI, our proposed network performed the best (SIWB: IoU 90.07%, F1_{score} 94.78%; and MSRWD: IoU 94.21%, F1_{score} 97.03%). Compared to previous CNN-based SOTA methods, our approach

TABLE III
ABLATION STUDY ON THE SIWB DATASET QUANTIFYING THE CONTRIBUTIONS OF THE SFS AND FFD COMPONENTS

Method	IoU _{water} (%)	F1-score _{water} (%)	Precision _{water} (%)	Recall _{water} (%)	OA (%)	Params (M)	FLOPs (G)
SDNet	87.99	93.64	95.77	91.61	97.05	7.28	6.57
SDNet-R-F	87.10	93.10	94.76	91.50	96.78	7.28	6.58
SDNet-S-M	85.59	92.24	94.54	90.05	96.40	9.79	8.60

TABLE IV
ABLATION STUDY ON THE MSRWD DATASET QUANTIFYING THE CONTRIBUTIONS OF THE SFS AND FFD COMPONENTS

Method	IoU _{water} (%)	F1-score _{water} (%)	Precision _{water} (%)	Recall _{water} (%)	OA (%)	Params (M)	FLOPs (G)
SDNet	93.04	96.43	97.24	95.64	98.65	7.28	6.57
SDNet-R-F	92.52	96.11	97.35	94.90	98.54	7.28	6.58
SDNet-S-M	91.76	95.70	96.67	94.75	98.38	9.79	8.60

TABLE V
QUANTITATIVE COMPARISON RESULTS AGAINST THE PRIOR SOTA METHOD ON SIWB

Method	IoU _{water} (%)	F1-score _{water} (%)	Precision _{water} (%)	Recall _{water} (%)	OA (%)	Params (M)	FLOPs (G)
DeeplabV3+	85.23	92.02	94.30	89.85	96.30	5.81	39.59
U2Net	84.20	91.43	92.32	90.55	95.96	1.13	51.21
SwinUnet	85.83	92.38	95.88	89.12	96.50	27.15	23.66
Crossformer	85.94	92.44	93.15	91.74	96.43	31.59	20.82
Segformer	86.09	92.53	96.31	89.03	96.58	13.49	1.43
Unetformer	87.04	93.07	94.93	91.28	96.77	11.72	11.66
CSwin	88.28	93.77	95.84	91.79	97.10	35.39	31.12
ConvNeXt	90.06	94.77	96.46	93.14	97.56	63.96	65.76
MSLANet	88.59	93.95	95.37	92.58	97.20	49.94	37.40
Waterformer	89.29	94.34	95.92	92.81	97.34	32.08	45.09
Seaformer	89.02	94.19	95.86	92.58	97.28	19.20	5.91
WatNet	89.42	94.41	95.72	93.14	97.40	31.18	33.61
SDNet	90.07	94.78	96.48	93.13	97.56	28.82	25.62

The bolded values indicate the best experimental results among the compared methods. Higher values represent better model performance.

TABLE VI
QUANTITATIVE COMPARISON RESULTS AGAINST THE PRIOR SOTA METHOD ON MSRWD

Method	IoU _{water} (%)	F1-score _{water} (%)	Precision _{water} (%)	Recall _{water} (%)	OA (%)	Params (M)	FLOPs (G)
DeeplabV3+	88.68	94.00	96.78	91.38	97.78	5.81	39.59
U2Net	89.36	94.38	95.51	93.27	95.96	1.13	51.21
SwinUnet	90.52	95.54	97.47	93.69	98.16	27.15	23.66
Crossformer	91.66	95.65	96.26	95.05	98.35	31.59	20.82
Segformer	92.49	96.10	97.48	94.79	98.53	13.49	1.43
Unetformer	92.50	96.10	96.77	95.44	98.54	11.72	11.66
CSwin	93.78	96.79	97.22	96.36	98.78	35.39	31.12
ConvNeXt	94.04	96.93	97.20	96.66	98.83	63.96	65.76
MSLANet	92.07	95.87	97.33	94.45	98.45	49.94	37.40
Waterformer	93.06	96.41	97.74	95.11	98.65	32.08	45.09
Seaformer	92.88	96.31	96.98	95.65	98.60	19.20	5.91
WatNet	93.57	96.68	97.56	95.82	98.75	31.18	33.61
SDNet	94.21	97.03	97.37	96.67	98.87	28.82	25.62

The bolded values indicate the best experimental results among the compared methods. Higher values represent better model performance.

delivers comparable accuracy while reducing the number of parameters and computational overhead by more than half. In addition, when compared to ViT-based methods, our model shows a 1–4% improvement in IoU and a 0.6–2.4% increase in F1_{score}, demonstrating superior adaptation to waterbody extraction,

regardless of large lakes or narrow rivers. Visual comparisons were presented in Fig. 10.

DeeplabV3+ and U2Net exhibit poor performance in identifying small circular waterbodies, often missing significant portions of the water regions. SwinUnet tends to misclassify areas

TABLE VII
COMPARATIVE EXPERIMENTS WITH THE TRADITIONAL METHOD NDWI ON
THE SIWB DATASET

Method	IoU _{water} (%)	F1score water (%)	Precision water (%)	Recall water (%)	OA (%)
NDWI- OTSU	49.84	61.43	50.92	77.41	77.40
NDWI- KMeans	47.19	58.98	46.24	81.41	73.67
SDNet	90.07	94.78	96.48	93.13	97.56

The bolded values indicate the best experimental results among the compared methods. Higher values represent better model performance.

TABLE VIII
COMPARATIVE EXPERIMENTS WITH THE TRADITIONAL METHOD NDWI ON
THE SIWB DATASET

Method	IoU _{water} (%)	F1score water (%)	Precision water (%)	Recall water (%)	OA (%)
NDWI- OTSU	24.81	44.25	32.76	68.14	65.83
NDWI- KMeans	24.58	42.46	31.19	66.48	64.15
SDNet	94.21	97.03	97.37	96.67	98.87

The bolded values indicate the best experimental results among the compared methods. Higher values represent better model performance.

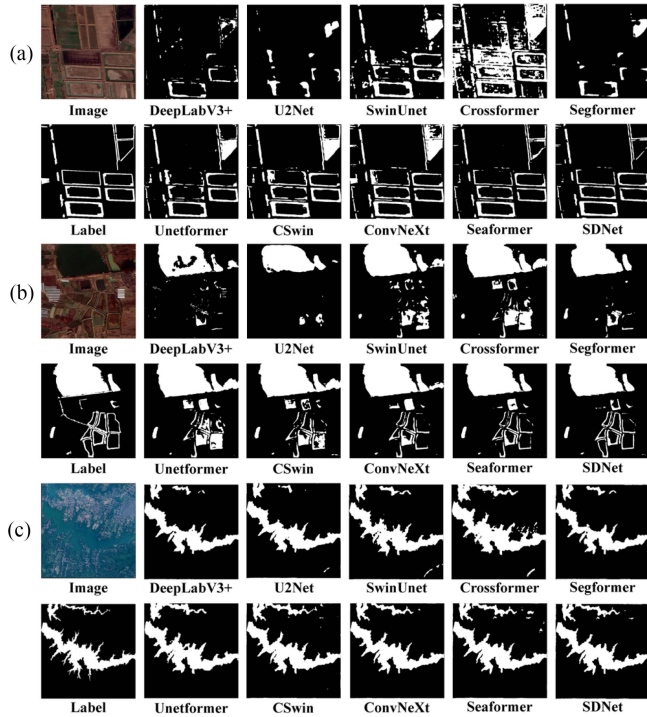


Fig. 11. (a)–(c) Visual comparison of WS on SIWB and MSRWD.

around circular waterbodies. Segformer demonstrates improved performance in capturing both large and small waterbodies. Unetformer, originally designed for urban RS segmentation tasks, also shows strong performance in waterbody extraction. While CSwin and ConvNeXt increase model parameters, enhancing generalization capabilities, this also results in a heavier computational burden. Conversely, our proposed SDNet offers a superior

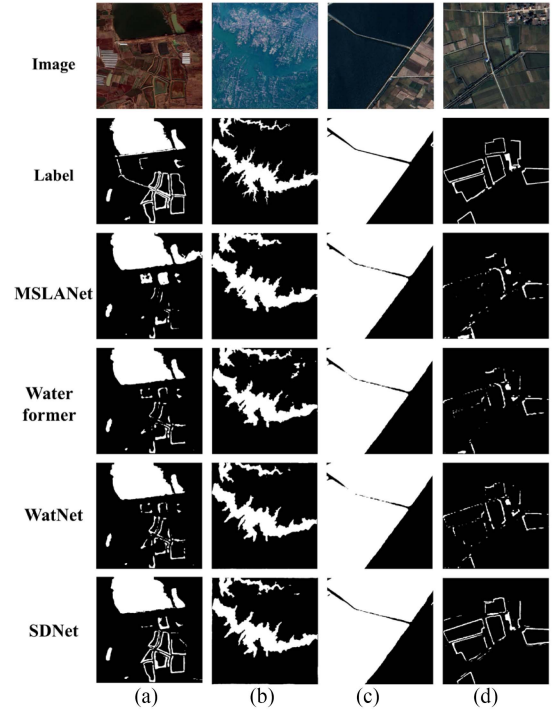


Fig. 12. (a)–(d) Visual comparison with advanced waterbody extraction methods on the SIWB and MSRWD datasets.

tradeoff between representational capacity and model dimensionality. Compared to advanced waterbody extraction methods such as MSLANet, WaterFormer, and WatNet, our proposed approach achieves superior performance in both qualitative and quantitative evaluations. Visual comparisons of representative cases are shown in Fig. 12.

In order to thoroughly assess the performance of SDNet and underscore its superiority over conventional approaches, we incorporated a comparison with the NDWI. The quantitative evaluation results are summarized in Tables VII and VIII.

In Tables VII and VIII, NDWI-OTSU refers to waterbody extraction using an automatically determined threshold via the OTSU method, while NDWI-KMeans denotes a clustering-based approach in which the cluster with the higher average NDWI value is identified as water. The experimental results demonstrate that our model significantly outperforms traditional approaches. Unlike spectral index methods that rely heavily on threshold selection, SDNet achieves superior segmentation accuracy and also exhibits faster processing speed. Visual comparison results are provided in Fig. 13.

The comparison with these traditional methods provides a more intuitive and comprehensive demonstration of the effectiveness of our proposed model.

2) *Comparative Experiments on the LoveDA Dataset:* LoveDA is a multiscale land cover semantic segmentation dataset that includes images from both urban and rural environments. We conducted further experiments to validate the advantages of SDNet for multiclass instance segmentation. The findings of the quantitative examination are displayed in

TABLE IX
QUANTITATIVE COMPARISON RESULTS ON LOVEDA WITH PRIOR SOTA MODELS

Method	Per-class							mPreci- sion (%)	mRecall (%)	mIoU (%)	OA (%)	Params (M)	FLOPs (G)
	BG	BL	RD	WT	BR	FR	AG						
DeeplabV3+	54.07	40.27	50.86	68.74	25.22	34.59	50.67	70.22	62.97	46.34	67.05	5.81	39.59
U2Net	47.55	37.77	46.37	60.61	33.08	38.18	48.21	63.25	61.92	44.54	64.00	1.13	51.21
SwinUnet	50.02	50.97	55.35	61.43	19.05	42.00	53.14	68.89	60.82	47.42	67.13	27.15	23.66
Crossformer	49.54	54.26	54.96	63.75	21.30	43.26	52.40	70.31	62.86	48.49	67.50	31.59	20.82
Segformer	46.38	57.01	50.49	65.22	17.72	39.59	49.30	66.65	61.25	46.53	64.75	13.49	1.43
Unetformer	51.41	55.84	50.72	64.39	20.70	41.23	49.61	70.26	60.64	47.66	67.20	11.72	11.66
CSwin	45.51	62.46	55.32	64.15	17.17	40.10	58.41	69.17	63.55	49.02	67.74	35.39	31.12
ConvNeXt	50.26	61.39	53.15	65.97	32.12	45.65	55.42	72.66	65.13	51.99	69.35	63.96	65.76
Seaformer	51.11	58.19	55.52	66.16	24.13	41.37	53.23	70.79	64.18	49.96	68.67	19.20	5.91
SDNet	53.14	65.80	56.48	71.13	28.55	42.18	53.34	75.45	65.07	52.94	70.38	28.82	25.62

BG: background; BL: building; RD: road; WT: water; BR: barren; FR: forest; AG: agriculture.

The bolded values indicate the best experimental results among the compared methods. Higher values represent better model performance.

TABLE X
QUANTITATIVE COMPARISON RESULTS ON UAVID WITH PRIOR SOTA MODELS

Method	Per-class								mPreci- sion (%)	mRecall (%)	mIoU (%)	OA (%)	Params (M)	FLOPs (G)
	CL	BL	RD	TR	VE	MC	SC	HM						
DeeplabV3+	54.56	87.67	67.08	74.87	65.43	66.65	54.66	35.04	78.29	75.92	63.25	83.87	5.81	39.59
U2Net	52.31	86.21	70.11	72.25	62.69	68.03	55.50	33.50	75.90	75.79	62.57	83.07	1.13	51.21
SwinUnet	62.44	90.64	76.37	76.53	67.46	70.98	62.08	31.72	79.77	79.52	67.28	86.35	27.15	23.66
Crossformer	60.03	91.41	74.12	77.34	67.96	68.42	60.85	39.04	79.77	79.87	67.42	86.93	31.59	20.82
Segformer	57.20	89.14	71.80	75.78	66.18	66.12	57.40	34.49	79.08	76.26	64.76	85.42	13.49	1.43
Unetformer	58.08	89.44	72.86	75.03	65.19	69.89	60.02	33.94	79.43	77.39	65.56	85.49	11.72	11.66
CSwin	63.43	91.81	72.93	77.94	69.72	73.17	65.18	37.85	82.22	79.71	69.00	87.52	35.39	31.12
ConvNeXt	65.47	92.36	78.02	77.86	69.74	72.78	66.96	42.13	82.34	81.93	70.67	88.14	63.96	65.76
Seaformer	61.76	90.94	75.95	77.04	67.31	72.35	66.67	41.90	81.77	80.40	69.24	87.00	19.20	5.91
SDNet	65.61	92.14	78.62	78.07	68.18	74.10	68.81	43.03	83.13	81.69	71.07	88.15	28.82	25.62

CL: clutter; BL: building; RD: road; TR: tree; VE: vegetation; MC: moving car; ST: static car; HM: human.

The bolded values indicate the best experimental results among the compared methods. Higher values represent better model performance.

Table IX. Notably, SDNet achieved the highest mIoU of 52.94% and OA of 70.38%, as well as the best recognition results in the waterbody category (IoU: 71.13%).

Visual comparison is illustrated in Fig. 14. DeeplabV3+ effectively classifies the background and waterbody categories effectively but often misclassifies farmland as forest. U2Net shows poor performance in identifying road categories. SwinUnet struggles with accurately extracting barren land. Segformer performs well in identifying buildings and waterbodies but makes numerous errors when detecting barren land. Unetformer exhibits minor errors across various categories. CSwin tends to merge surrounding background areas into other classes, leading to misclassification.

In contrast, our proposed SDNet demonstrates exceptional performance in the extraction of buildings, roads, and waterbodies. The excellent pixel-level recognition result comes from the comprehensive consideration in the model's design: selective SA mechanism is utilized to identify remote associations, and features from many stages are merged inside the multibranch modeling approach.

3) *Comparative Experiments on the UAVid Dataset:* UAVid is a large-resolution suburb-scene dataset for a pixel-level segmentation task, distinguished by deep annotations, varied categories, and considerable obstacles. Our proposed SDNet achieved the best segmentation performance, with mIoU of 71.07% and OA of 88.15%.

Detailed comparative results of evaluation metrics are presented in Table X. Compared to convolution-based neural networks such as DeeplabV3+ and ConvNeXt, our method demonstrates superior expressive capability with fewer parameters, attributable to the STPF module that establishes various recognition patterns. For ViT-based networks such as SwinUnet and CSwin, we implemented a TSSA mechanism to enhance model efficiency. Our approach falls within the CNN-ViT hybrid network category, akin to Unetformer, yet it achieves better recognition results.

Fig. 15 illustrates the comparative recognition effectiveness of each network. SDNet achieves strong performance for both small-scale pedestrians and cars, as well as large-scale roadways and structures.

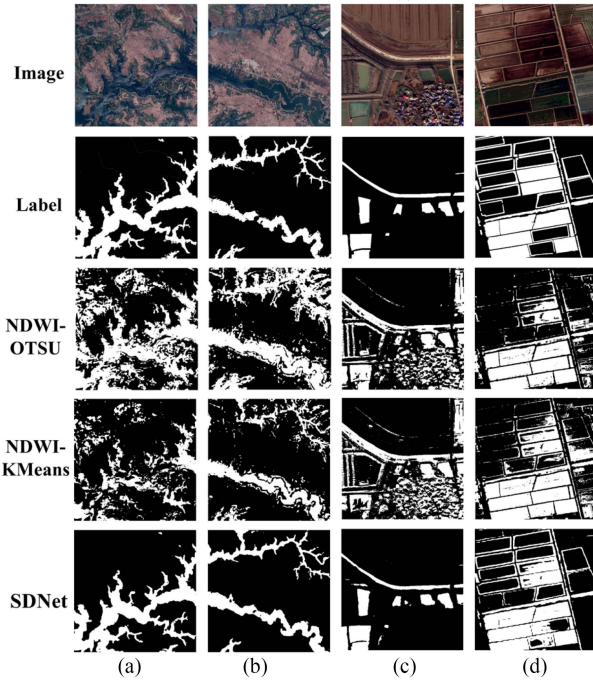


Fig. 13. (a)–(d) Comparison of segmentation results between SDNet and the traditional NDWI approach on SIWB and MSRWD datasets.

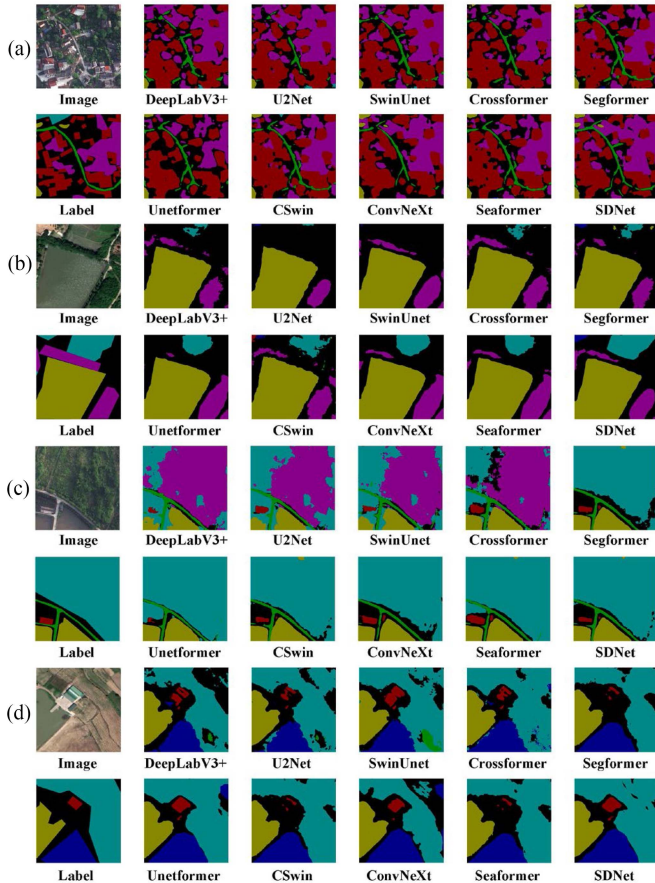


Fig. 14. (a)–(d) Visual comparison of multiclass instance segmentation results on LoveDA.

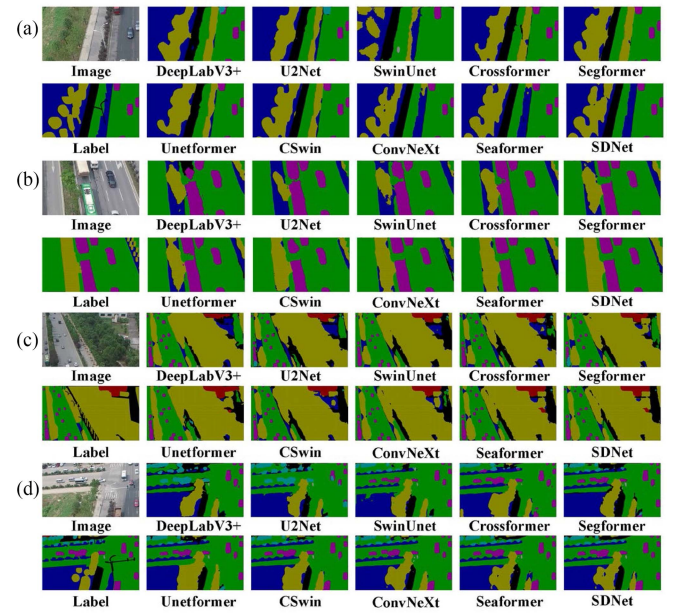


Fig. 15. (a)–(d) Visual comparison of multiclass instance segmentation results on UAVid.

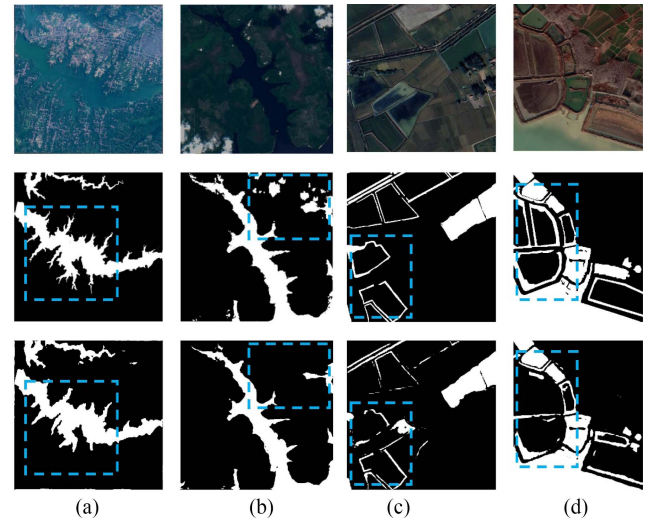


Fig. 16. (a)–(d) Segmentation weakness on SIWB and MSRWD. The top row represents four input RGB images. The middle row denotes the corresponding GT. The bottom row shows the WS results produced by our SDNet.

4) *Discussion About Limitations of SDNet*: Despite the effectiveness of our method demonstrated across SIWB, MSRWD, LoveDA, and UAVid datasets, there remain some limitations. In Fig. 16(a), fine details of waterbodies are overlooked; in Fig. 16(b), some water areas obscured by clouds are neglected; in Fig. 16(c), small circular waterbodies are not accurately identified; and in Fig. 16(d), elongated waterbodies are missed.

The aforementioned limitations are challenging to surmount, yet we can continue to explore them further. We have envisaged many avenues for future endeavors to rectify these deficiencies.

1) *Hardware Constraints*: Limited computational resources prompted us to focus primarily on the decoder. However, there are additional innovative techniques that can achieve

strong performance under low computational conditions, such as unsupervised domain adaptation, zero-shot deep learning, and so on. In the future, we will explore the integration of these methods to achieve better results with more lightweight models. These small plug-in modules should also be easily integrated into large-scale RS models.

- 2) *Data Processing Constraints*: In many previous studies, most AI models have primarily processed only the RGB channels from multispectral imagery, with few works incorporating the NIR channel. This practice leads to the loss of significant hidden information that is invisible to the human eye, effectively reducing RS images to natural image processing. In future work, we aim to explore preprocessing techniques to minimize information loss.

V. CONCLUSION

In this article, we propose SDNet for WS in RS imagery. SDNet is composed of a downsampling feature aggregation branch and an upsampling semantic reconstruction branch. The backbone network for feature extraction is interchangeable, providing our model with excellent scalability. Our primary focus is on optimizing the decoding process, where the core unit is the sandwich decoder.

Each sandwich decoder consists of three layers: 1) the bottom layer employs a TSSA strategy to capture long-range dependencies and reduce redundant computations; 2) the middle layer, derived from the original image feature encoding, contains rich information and aligns the feature channels; and 3) the top layer is an STPF module, which integrates multiscale information to enhance segmentation performance. Extensive ablation studies on waterbody datasets validate the effectiveness of each component. Furthermore, we conduct comparisons with SOTA algorithms on multiclass instance segmentation datasets, demonstrating the robustness and superiority of SDNet.

For future work, we plan to explore lightweight plug-ins for large RS models to enhance performance in engineering tasks. Considering the distinctions between remotely sensed and natural images, we also aim to develop neural networks specifically tailored to the RS domain for practical applications. We hope to develop small plug-ins for various applications that facilitate the RS large model and are simple to construct.

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